Testing SLAM when vision fails

A reproducible benchmark for visual–inertial tracking.

Inspired by MM3DGS SLAM
Sun et al., 2024



Original procedural studio · no downloaded scene assets

Goal: compare tracking through weak or missing visual observations.

A reproducible visual–inertial testbed

Procedural studio · synchronized sensors · ground truth reserved for scoring

24 s
smooth 6-DoF sequence

30 Hz
RGB + metric depth

100 Hz
noisy, biased IMU

1 MuJoCo rig
Prescribed motion
Known camera–IMU calibration

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2 Sensor streams
RGB, depth and raw IMU
Replay the same data

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3 Estimator + scoring
Official-code data bridge
ATE, RPE and pose coverage

THREE MATCHED VISUAL CONDITIONS

Rich
normal visual reference

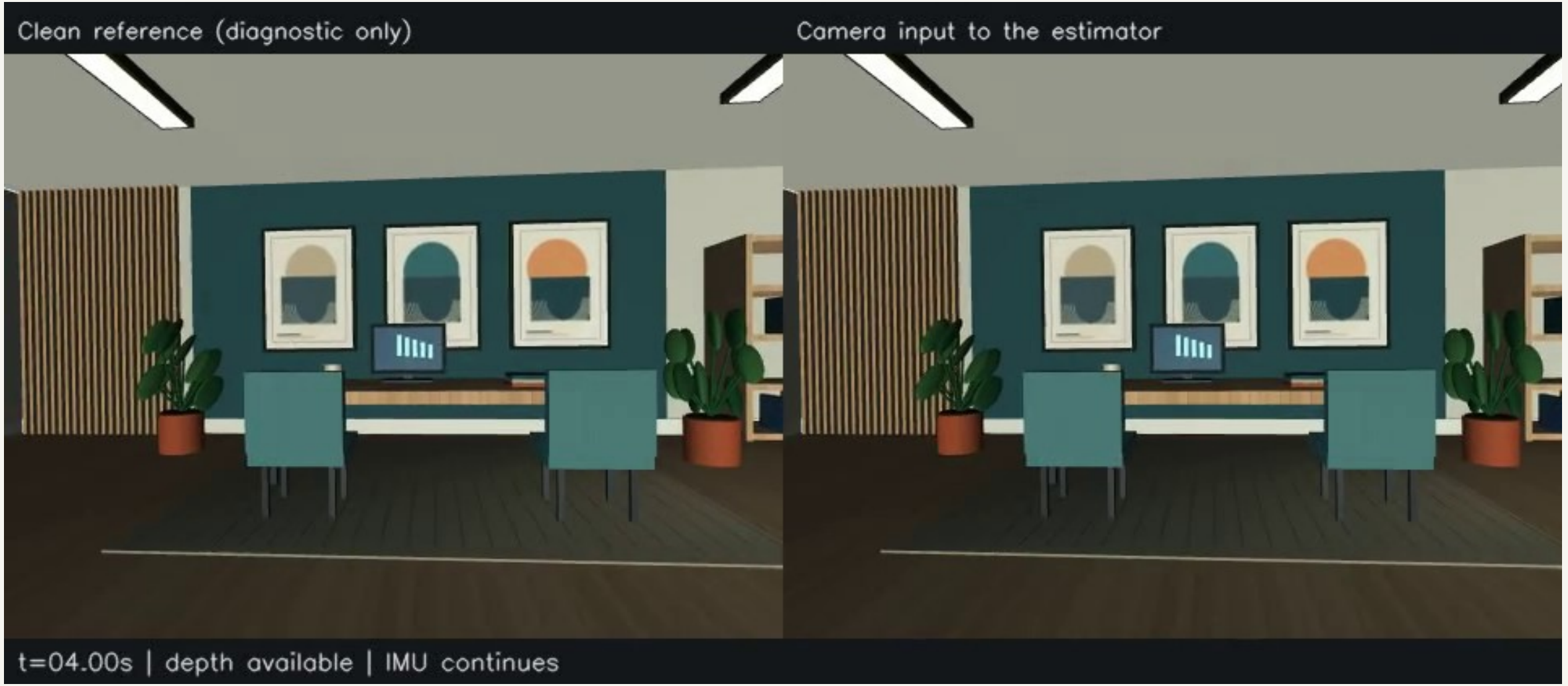
Low texture
dimming + blur

Blackout
RGB and depth unavailable

Demo: four seconds without vision

Open MP4 fallback

Left: clean reference only. Right: actual camera input. Click the video to play.



0–8 s: normal observations | 8–12 s: blackout | 12–24 s: visual recovery

Verified simulator. Estimator comparison comes next.

The current evidence validates the testbed and the sensor streams.

VERIFIED

30 checks passed

Physical IMU output and calibration
Camera/depth geometry and timing
All 120 blackout frames invalid

SANITY CHECK / 4-SECOND OUTAGE

Predictor	Endpoint error
Constant velocity	4.44 m
Noisy IMU propagation	0.051 m

Oracle starting pose + velocity · one seed · sensor check, not SLAM accuracy

Next: visual-only → IMU initialization → tightly coupled fusion

Run full MM3DGS on CUDA; compare error, coverage and recovery across seeds.
The full SLAM pipeline has not run here; necessity of tight coupling remains unproven.